

NORDENSKIOLD BASE, Antarctica

WMO: 890140

Lat: 73.050S

Lon: 13.383W

Elev: 1631

StdP: 13.85

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) -30.3	(c) -26.0	(d) -44.3	(e) 0.4	(f) -12.6	(g) -40.3	(h) 0.6	(i) -16.3	(j) 81.4	(k) 4.9	(l) 75.8	(m) 4.4	(n) 15.3	(o) 30	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 7.6	(c) 31.7	(d) 26.8	(e) 29.4	(f) 25.3	(g) 27.6	(h) 24.1	(i) 27.4	(j) 30.4	(k) 25.9	(l) 28.6	(m) 24.7	(n) 27.1	(o) 12.4	(p) 50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 24.8	(b) 20.0	(c) 26.7	(d) 23.1	(e) 18.4	(f) 25.7	(g) 21.4	(h) 17.1	(i) 24.6	(j) 10.2	(k) 30.6	(l) 9.6	(m) 28.7	(n) 9.1	(o) 27.3	(p) 32.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 78.7	(b) 67.7	(c) 56.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	22.0	15.8	7.5	1.3	-1.0	-4.6	-7.6	-7.5	-5.6	0.8	13.6	21.1		
	DBStd	12.96	3.88	5.59	6.48	8.18	9.15	8.69	8.67	10.15	8.65	8.03	5.62	3.97		
	HDD50	16571	868	957	1317	1462	1580	1638	1786	1782	1667	1526	1093	895		
	HDD65	22046	1333	1377	1782	1912	2045	2088	2251	2247	2117	1991	1543	1360		
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	19.9	15.2	17.3	20.3	19.9	24.7	21.2	20.4	18.8	23.5	18.3	22.1	16.6		
	PrecAvg	18.2	1.1	1.0	1.3	1.3	1.7	1.4	1.4	1.7	1.7	1.9	2.2	1.4		
(15) Precipitation	PrecMax	28.7	3.0	3.0	4.2	3.0	3.9	3.9	3.7	3.4	4.7	3.9	5.3	3.1		
	PrecMin	8.6	0.2	0.1	0.3	0.2	0.4	0.2	0.2	0.3	0.2	0.2	0.5	0.3		
	PrecStd	5.7	0.7	0.7	0.8	0.8	0.9	0.9	0.9	0.8	1.1	1.0	1.2	0.8		
	PrecStd	5.7	0.7	0.7	0.8	0.8	0.9	0.9	0.9	0.8	1.1	1.0	1.2	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	33.6	23.4	20.8	20.2	17.6	13.2	20.5	13.6	20.0	27.5	34.1		
		MCWB	29.2	27.8	22.0	18.6	19.5	15.2	10.7	18.3	10.4	16.8	23.7	27.2		
	2%	DB	31.9	27.5	20.6	17.1	17.6	13.2	10.0	16.0	10.7	16.9	24.8	31.5		
		MCWB	27.6	23.5	19.2	15.8	16.4	11.7	8.4	14.3	8.5	14.6	21.2	26.2		
	5%	DB	29.7	25.3	18.8	15.0	14.9	10.7	7.7	11.2	8.9	14.2	22.3	29.4		
		MCWB	26.0	22.4	17.2	13.4	13.4	8.9	6.5	9.2	7.3	11.9	19.9	24.9		
	10%	DB	28.0	23.5	16.9	12.8	12.1	8.4	5.1	6.2	6.9	11.6	20.8	27.4		
		MCWB	24.6	21.0	15.4	11.3	10.6	6.8	3.6	4.5	5.7	9.8	18.7	23.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.1	26.7	22.7	19.7	20.1	14.8	10.5	19.8	11.4	18.4	25.2	29.3		
		MCDB	33.2	28.4	22.9	20.1	20.1	16.2	11.9	20.8	13.3	19.4	27.3	32.9		
	2%	WB	28.0	24.3	19.6	16.5	17.2	11.6	8.3	14.6	8.8	15.0	22.2	26.9		
		MCDB	31.1	26.0	20.4	17.3	17.7	12.5	9.3	16.2	10.2	16.9	23.8	30.3		
	5%	WB	26.5	22.7	17.6	13.7	14.2	8.6	5.7	9.7	7.2	12.3	20.7	25.5		
		MCDB	29.0	24.5	18.6	14.7	15.2	9.7	6.7	11.7	8.5	14.0	22.1	28.8		
	10%	WB	25.2	21.2	15.6	11.5	11.0	6.0	2.5	5.5	5.3	10.1	19.2	24.3		
		MCDB	27.4	23.0	16.9	12.8	12.5	7.7	3.8	6.8	6.6	11.7	20.9	27.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	7.7	8.0	8.9	9.3	9.8	10.5	10.0	9.3	8.9	7.9	7.5		
		MCDBR	8.9	8.8	7.5	9.4	8.6	9.6	10.7	10.9	9.0	9.4	8.9	9.6		
		MCWBR	7.0	7.5	7.4	8.7	8.1	8.8	11.1	10.0	8.4	8.2	7.9	7.4		
		MCDBR	8.5	8.1	7.1	8.8	7.7	8.9	11.3	10.9	8.7	9.0	8.2	8.7		
	5% WB	MCWBR	7.0	7.3	7.2	8.6	7.7	8.5	10.9	10.2	8.0	8.2	7.4	7.0		
(40) Clear-Sky Solar Irradiance	taub	0.231	0.215	0.187	0.115	N/A	N/A	N/A	N/A	0.112	0.174	0.197	0.214	0.230		
	taud	2.203	2.225	2.211	1.910	N/A	N/A	N/A	N/A	1.876	2.152	2.182	2.200	2.190		
	Ebn at Noon	318	302	269	176	0	0	0	0	170	270	308	324	324		
	Edh at Noon	35	29	19	7	0	0	0	0	7	20	30	35	38		
(44) All-Sky Solar Radiation	RadAvg	2601	1663	708	134	0	0	0	0	46	449	1293	2352	2984		
	RadStd	68	49	26	9	0	0	0	0	3	25	52	63	70		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page